

Designing High-Impact Experiments for Human–Autonomy / AI Teaming

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Abstract

The potential to create autonomous teammates that work alongside humans has increased with continued advancements in AI and autonomous technology. Research in human–AI teams and human–autonomy teams (HATs) has seen an influx of new and diverse researchers from human factors, computing, and teamwork, yielding one of the most interdisciplinary domains in modern research. However, the HAT domain's interdisciplinary nature can make the design of research, especially experiments, more complex, and new researchers may not fully grasp the numerous decisions required to perform high-impact HAT research. To aid researchers in designing high-impact experiments, this article itemizes four initial decision points needed to form a HAT experiment: deciding on a research question, deciding on a team composition, deciding on a research environment, and deciding on data collection. For each decision point, this article discusses these decisions in practice, providing related works to guide researchers toward different options available to them. These decision points are then synthesized through actionable recommendations to guide future researchers. The contribution of this article will increase the impact and knowledge of HAT experiments.

Keywords

human–autonomy teams, experiments, methods, measurement, synthetic task environments

Introduction

Research in the field of human–AI teams or human–autonomy teams (HATs) has experienced rapid growth in tandem with the advancement of artificial intelligence (AI) technologies, leading to the increasingly plausible implementation of HATs in the real-world (Berretta et al., 2023; O'Neill et al., 2022). These HATs are defined by the interdependent collaboration of humans and artificial teammates with a high degree of autonomy in order to complete a shared goal, which is often accomplished through AI

integration (O'Neill et al., 2022). Positioned at the intersection of human factors (Lyons et al., 2021; O'Neill et al., 2022), teamwork (O'Neill et al., 2023a), macrocognition (Klein et al., 2003; Patterson et al., 2018), cognitive systems engineering (Dalal & Kasper, 1994; Johnson et al., 2019), and computational AI research (Bansal, Nushi, Kamar, Lasecki, et al., 2019; Bansal, Nushi, Kamar, Weld, et al., 2019), the HAT domain has cultivated a diverse and continually expanding body of research. Consequently,

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Correction (December 2025): This article has been updated to correct two references in which the name of the first author, Cooke, N.J., was mistakenly omitted. Both references have now been amended accordingly.

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performing research within the HAT domain is complicated by a myriad of decisions, with the breadth of the domain making researchers potentially unaware of all of the options available to them.

In particular, experiments in the HAT domain are especially impacted by this complexity. With experiments having to constantly consider causal relationships, hypotheses, control variables, measurement, and external validity (Kirk, 2009), experimental design can be seen as a chain of interconnected decisions that require consistency. Especially in the HAT domain, experimental creation can be difficult as it needs to maintain considerations of human teammates, autonomous teammates, teams as a whole, and a team's environment (N. Cooke et al., 2020), each of which is continuously changed due to technology advancements. In turn, existing and future researchers need an understanding of the decisions that lead to the creation, performance, and publication of high-impact experimental HAT research.

The convergence of multiple disciplines presents itself as both advantageous and daunting for experimental HAT research. Historically, HAT research was more often focused on demonstrating and explaining the possible viability of HATs and autonomous teammates (Chen, 2018; McNeese et al., 2018; Shively et al., 2018; Tokadli & Dorneich, 2019). Over time, experiments have expanded to explore and optimize different characteristics of HATs. Most notably, the computational aspects of autonomous teammates (Bansal, Nushi, Kamar, Lasecki, et al., 2019; Bansal, Wu, et al., 2021), the behavior of autonomous teammates (Flathmann, Schelble, Rosopa, et al., 2023; McNeese, Schelble, et al., 2021; R. Zhang et al., 2023), the joint cognitive design of autonomous teammates (Hollnagel & Woods, 1983; Woods, 1985; Woods & Patterson, 2001), behavior of other human teammates (Habibian & Losey, 2022; Lakhmani et al., 2022; Schelble, Flathmann, et al., 2022), and the formation of macrocognition (Klein et al., 2003, 2007) are some of many important considerations to advancing HAT research. Due to this breadth, the contributions of HAT experiments have led to rapid growth and interest within the domain, but various different approaches exist to performing these experiments.

For instance, HAT researchers have adopted diverse platforms, including team-specific research

platforms (Demir, Likens, et al., 2019; Demir, McNeese, & Cooke, 2019; McNeese et al., 2018), common video game platforms (Rosero et al., 2021; Schelble, Lopez, et al., 2022), physical autonomous teammates (Alarcon et al., 2024), and online survey platforms for conducting experiments (Lyons et al., 2023; Textor et al., 2022; R. Zhang, Flathmann, et al., 2024). Similarly, the operationalization of autonomous teammates within HAT experiments varies, ranging from experimenters simulating autonomous teammates (Demir et al., 2017; McNeese et al., 2018) to the use of real AI technologies (Bansal, Wu, et al., 2021) or pre-recorded demonstrations (Lyons et al., 2023; R. Zhang, Flathmann, et al., 2024). Additionally, the dependent variables of interest to HAT research have expanded, exemplified by the numerous outcomes HATs can impact, such as trust (Luo et al., 2022; McNeese, Demir, et al., 2021), performance (Bogg et al., 2021; Munyaka et al., 2023), workload (Flathmann, Schelble, Rosopa, et al., 2023; Hemmer et al., 2023), or shared understanding (Andrews et al., 2023; Schelble, Flathmann, et al., 2022). With the HAT domain expanding, researchers will need a starting place to approach this breadth and design a high-impact experiment.

By leveraging supporting HAT literature, this article provides an approachable foundation on the experiment creation process in HATs, which can be used for future experimental design, systematic reviews, and research assessment in HATs. To do so, this article has worked to walk researchers through the HAT experiment design process, highlighting the common decisions researchers need to make to design a high-impact HAT study. In turn, this article extracts design considerations for four foundational processes that will provide experienced and new HAT researchers with an initial guidepost to developing experiments: (1) formulating a research question, (2) determining team composition, (3) selecting a research task and platform, and (4) deciding on data collection methods. To supplement this discussion, three case studies have also been compiled and provided in an accompanying appendix, which will highlight how these design considerations can manifest into a high-impact HAT research study.

This article contributes to the HAT research field in three important and timely ways. First, this article provides a resource uniquely made for

the HAT domain. As the domain continues to advance via empirical research, this article will serve as an important resource for teaching new researchers and streamlining experimental design. Second, while this work is not a systematic review itself, the content of this article provides a structure through which HAT experiments can be approached and reviewed. As such, this article will improve the structure of future HAT experiments and guide future systematic reviews on specific experimental decisions, such as team composition, tasks, or measurement. Finally, this article starts an introspective discussion on HAT research and experiments, which need to be critically assessed to ensure HAT research provides a strong empirical foundation. Consequently, this article not only encourages but guides the creation of similar meta-pieces, such as systematic reviews on experimental methods and measurement, further enhancing the capabilities of the HAT domain and its researchers.

Deciding on a Research Question

For the first decision point, this article highlights the importance of the research question, which discusses development in general and unique considerations in the HAT community. This article will follow an input, process, and output (IPO) framework for teaming (Landy & Conte, 2016), which has been recently applied to HAT research (O'Neill et al., 2023b). The development of HAT research questions often identifies specific factors of a team's environment, AI design, composition, leadership, or task (input); interactions, decisions, and team constructs (process); and resulting impacts on the team's environment and effectiveness (output). Notably, a singular factor (e.g., trust) can serve as an input, process, and output in HATs. As such, this section does not specifically identify what specific factors should be inputs, processes, and outputs but rather how such factors can be interwoven through research questions, which will help guide all subsequent decisions made in a HAT experiment.

HAT Processes and Outputs

The IPO framework is noteworthy as a cyclical framework of effectiveness derived from decades

of teamwork and technology-based research (Mathieu et al., 2008). Research into "Processes and Outputs" have resulted in several core teamwork constructs being defined in human-human teams, such as team cognition (Cooke et al., 2013; Salas & Fiore, 2004), awareness (Gutwin & Greenberg, 2004; Hinds & Kiesler, 2002), backup behaviors (Barnes et al., 2008; Salas et al., 2005), communication (Marlow et al., 2018; Salas et al., 2008), and transaction memory (Wegner, 1987; Yoo & Kanawattanachai, 2001) (see case studies for examples). Outside of the teaming field, computer science, human factors, and computer-supported cooperative work have also identified core processes and outputs for AI technology. For example, collaborative performance (Grace et al., 2018; Topol, 2019), intent to use (Flavián et al., 2021; Liu et al., 2022), satisfaction (Ashfaq et al., 2020; Brill et al., 2022), and usefulness (Kelly et al., 2023; Kim et al., 2021) are all seen as important to HATs. As mentioned, many of these processes are not unique to HATs, as technology and teaming perceptions are common in automation and teamwork research; however, their operationalization within HAT research is going to differ. The introduction of new AI technologies, such as Large Language Models has enabled autonomous teammates to hold both social and functional roles in teams (Sidji et al., 2024; Zhang et al., 2024). These roles persist through various autonomous systems, ranging from robotics to vehicles to digital assistants. While this range in applicability improves accessibility, it also consequentially obfuscates human awareness in collaborative decision-making as these systems become inherently less transparent and observable to humans (Barman et al., 2024; Sarker, 2024). These changes, among others, mean that processes and outputs in HATs mean that people's perceptions in interactions with HATs look different from automation or basic interaction. HAT experiments need to build on these prior works to uniquely define these processes in HATs, design autonomous teammates around them, and evaluate these teammates through additional experiments.

Inputs to HAT Research Questions

One of the more unique considerations of HAT research is that inputs can be unique to

autonomous teammates, unique to human teammates, or shared between humans and autonomous teammates. As such, prior research has heavily explored the behavior of autonomous teammates or the composition of a HAT (see case studies for examples). Additionally, the design of autonomous teammate ethics (B. G. Schelble, Lopez, et al., 2022; Textor et al., 2022), etiquette (Flathmann, McNeese, et al., 2023), explanations (Ling et al., 2023; Taesiri et al., 2022), transparency (Chen & Barnes, 2023; Chen et al., 2016; Luo et al., 2022), and predictability (Lyons et al., 2023) can impact the perceptions humans form for autonomous teammates and teams. Again, these parameters are often not unique to HATs, as automation has explored several of these technical inputs and social inputs have always been important to teaming research. However, HATs and autonomous teammates can change the nature of these inputs in two ways that need to be accounted for. First, the increased capability and autonomy mean that these inputs will change from prior work, as the amount of inputs AI can provide teams will increase. Second, continued advancements in AI technology will (1) change the importance of these inputs to humans and (2) allow AI and autonomous teammates to consider these inputs in real time, meaning that autonomous and human teammates can both adapt around team processes in HATs (Andrews et al., 2023; Scheutz et al., 2017). For example, autonomous teammates would benefit from being able to predict human workload to improve its own behavior and teamwork (Giolando & Adams, 2024). These changes will fundamentally shift the amount and role of teaming inputs in HATs, as many inputs will need to be recontextualized from prior work and used as design features in autonomous teammates.

Synthesizing Recommendations for Research Question Selection

Based on insights taken from the current HAT landscape, HAT-specific recommendations can be made to improve the formation of research questions. The following recommendations denote criteria that researchers can use to leverage this model to create high-impact research questions explicitly designed for HATs. The following list is not exhaustive, and continued innovation as well

as niche cases in HAT research will require researchers to make their own decisions.

- Explore prior research in teamwork and technology to identify inputs, processes, and outputs of interest to HATs, but identify how those factors are likely to change due to increased levels of autonomy, AI complexity, and teamwork found in HATs.
- View HAT processes and outputs as potential inputs for research questions and also view them as inputs for the design and parameters of AI and autonomous teammates.
- Leverage HAT research questions to explore novel teamwork processes in HATs, recontextualize existing teaming factors in HATs, or evaluate autonomous teammate technologies.

Deciding on a Team Composition

Alongside determining a research question, researchers must also decide on the composition of HATs within their experiments. Based on the research question, team composition can be a fundamental place to operationalize the inputs and processes based on the researcher's interest. Thus, deciding on team compositions requires constant consideration of interdependence, the unique capabilities of human and autonomous teammates, and the unique impact of AI and autonomy on team dynamics with great particularity.

Deciding on Interdependent Team Roles for Human and Autonomous Teammates

This article can begin to explore the selection of a HAT composition with selecting team roles for human and autonomous teammates. Researchers must focus on creating interdependence through these roles to make this selection. At its core, team interdependence refers to the division of team responsibilities while maintaining a joint cognitive perspective on these responsibilities (Hoffman et al., 2002). In HATs, interdependence requires that teams' roles merge the cognitive and behavioral abilities of humans and autonomous teammates (Bradshaw et al., 2013; Johnson et al., 2019). An approachable way to outline interdependence in HATs is through the operationalization of roles, which defines the responsibilities of

teammates and how joint cognition and effort are required to complete a team task (Aubé & Rousseau, 2005; Beauchamp & Bray, 2001).

What separates HATs from prior constructs like automation or human-AI interaction is the increased overlap and distinction between humans and AI. Indeed, with Large Language Models and other AI technologies advancing, AI is becoming an increasingly capable substitute for humans in many instances. However, AI also presents unmatched capabilities compared to any other prior iteration of intelligent technology, meaning it can do things humans cannot. In turn, team roles can either explore joint cognitive compositions, where humans and AI leverage their overlaps to enhance teamwork (as seen in Case Study 3), or functional interdependence can be used to identify the most complementary roles for AI and humans (as seen in Case Study 1 and 2). This means that the number of roles available to AI and the roles that humans will ideally occupy will enormously differ from prior automation, teaming, and interaction research. Researchers must consider the growing overlap and distinction between humans and AI technology to select the best roles and composition for their research question.

Unique Decisions for Autonomous Teammate Implementation

Beyond shared team roles, several unique decisions for autonomous teammates in HAT experiments also exist.

Digital and Physical Autonomous Teammates. First, growing trends in AI technology emphasize that AI and autonomy can manifest both digitally and physically. While a large majority of HAT research is conducted in the digital landscape (McNeese et al., 2018), the growing number of autonomous robotic platforms is making physical HAT research much more feasible within the research domain (von Ellenrieder et al., 2022). However, choosing between these two options is fairly straightforward, as research leveraging robotics should likely only be pursued if physical considerations are central to one's research questions. Indeed, the physical presence and commercial nature of robotic platforms could confound participants' perceptions (Schwind et al., 2018) and create potential

concerns around physical safety amongst participants (Vasic & Billard, 2013; Zacharaki et al., 2020). The growing breadth of commercial autonomous technologies means that this decision will become more complex and important over time, as each system will carry a unique modality, interaction paradigm, and capability.

Autonomous Teammate Implementation. Regardless of whether an autonomous teammate is physical or digital, these teammates need a way of interacting and being controlled within an HAT. Notably, two choices present themselves to researchers: the Wizard of Oz (WoZ) approach or real AI systems. Traditionally, the limited and complex nature of AI and autonomy led a large portion of HAT, human-automation, and human-AI interaction research to use WoZ techniques, which have human experimenters pose as autonomous and AI technologies (Dahlbäck et al., 1993). However, recent advancements in AI mean that autonomous teammates powered by real AI technology are more feasible (McNeese et al., 2023). Indeed, HATs present an opportunity to leverage a growing range of commercial or open-source AI technologies, with prior work often prioritizing the uses of WoZ (N. Cooke et al., 2020). However, unlike prior work in automation, adopting these technologies will carry inherent uncertainty and risk due to AI technologies' often propriety, black-box, or highly computational nature (Trusilo, 2023). Yet, real AI technologies could increase the validity of HAT research and create a direct line for design recommendations to real AI technologies. This would be especially true for multi-autonomy teams, where autonomous teammates will need to interact with one another, which would look significantly different than human-human or human-autonomy interaction (Adams, 2009; Vig & Adams, 2006). As such, research needs to continuously consider the utility of new AI technologies and leverage them, knowing that WoZ techniques can always serve as a possible fallback.

The Number of Autonomous Teammates. With the growing prevalence of AI technology, real-world HATs are expected to leverage multiple autonomous teammates in the future (Hunt et al., 2023; Steane et al., 2023). However, more autonomous teammates do not equate to universal HAT

performance and perception benefits. Numerous autonomous teammates may even harm HAT outcomes due to increased cognitive workload (Hussein et al., 2022). While these harms and impacts may be of consideration for experiments, they may confound experimental results where multiple autonomous teammates are not needed. Research studies explicitly interested in how human workload can handle interactions with multiple autonomous teammates will need to incorporate multiple autonomous teammates (Manjunatha et al., 2020). Contrarily, research that is more focused on intimate human-AI relationships would likely want to isolate experimental results to a singular AI technology. Based on existing research in HATs, the ideal way to approach the number of autonomous teammates is based on the ratio of autonomous teammates to human teammates (Musick et al., 2021). When this ratio favors autonomous teammates, team processes and outcomes can be negatively impacted, so a human-favoring ratio is likely ideal in most cases.

Unique Decisions for Human Teammates Implementation

With team roles assigned and autonomous teammates selected, final team composition decisions can turn toward the unique decisions needed to place human teammates in HATs.

The Number of Human Teammates. If one's research question focuses more on human-AI outcomes and the human-AI relationship, then using only one human teammate may be ideal for isolating these relationships. For example, prior works have isolated a human-AI relationship by only including one human and one AI in the experiment (Lyons et al., 2023). However, research questions more concerned with human-human outcomes may need to leverage more human teammates to explore human-human dynamics and differences (Textor et al., 2022). This decision becomes increasingly important for HATs as the social nature of AI technology and autonomous teammates are already changing human-human relationships.

The Use of Human Confederates. Researchers are also faced with whether human confederates are needed. Human confederates can serve essential roles in experiments, as they control human actors'

behavior within teams (Frost, 2019). For example, Lyons et al. (2023) needed to explore autonomous teammates that act against the expected behavior of a human teammate (see Case Study 2). Within HATs, confederates have become necessary when exploring the sharing of perceptions for autonomous teammates and how human-human interaction impacts human-autonomy interaction (Demir, Likens, et al., 2019). Explicit considerations should be made of how confederates and autonomous teammates should interact since the research team would mostly control these interactions.

Synthesizing Recommendations for Team Composition Selection

Similar to research questions, the above recommendations and options can be synthesized to provide future researchers with quick guidance on selecting a team composition for their experiment. Thus, the following items should provide general guidance for forming team compositions in HAT experiments:

- To prevent autonomous teammates from overwhelming HATs, use the minimum required number of autonomous teammates in a HAT to answer your research question.
- If a research question is not explicitly concerned with physical autonomous teammates, consider using digital autonomous teammates to reduce confounds with robotic design.
- When possible, use real AI teammates to increase the applicability of research to modern HATs but remember that a WoZ can be a fallback if an AI is not available or possible.
- Consider having multiple humans to explore human-human relationships in HATs, but only use confederates when necessary to more accurately explore these relationships.

Deciding on a Research Task and Platform

Building on an identified team composition, researchers can operationalize and contextualize their research questions and compositions by selecting an experiment task environment and platform. For HAT experiments, tasks comprise of

teams working in teamwork-specific Synthetic Task Environments (STEs), adapted video game platforms, or physical environments.

Deciding to Use Digital Task Environments

One can begin to decide on their ideal research task for an experiment, which first requires researchers to determine whether their experiment should leverage a digital or physical environment. HAT research commonly leverages digital STEs, which simulate an existing real-world task in a controllable and safe manner (Gray, 2002). STEs are not exclusive to HAT research, as they have a rich history in general research (Gray, 2002) and especially team research (Cooke et al., 1999; Cooke & Shope, 2004). For HATs, the greatest benefit of most STEs is that they allow participants to collaborate and complete an actual task together, as many people may have never completed a task with an autonomous teammate. For HAT research specifically, STEs often manifest as either team-specific research platforms or team-based video game platforms repurposed for research.

Digital HAT-Specific STEs. Team STEs present research-oriented solutions within team-specific research environments that can immediately accommodate and emphasize teamwork during an experiment (Demir et al., 2015; Cooke & Shope, 2005). Within this category, numerous platforms have gained traction in HAT research, such as the CERTT UAV STE (Cooke & Shope, 2004), NeoCities (McNeese et al., 2005; Schelble et al., 2020), and Blocks World for Teams (Jensen, 2021; Johnson et al., 2009). In turn, HAT members within these environments can form bonds centered around teamwork and team tasks. Similarly, the research-specific use of these environments means that they are often already validated by various research studies, with the CERTT UAV STE being a cornerstone of teamwork research (Demir, McNeese, & Cooke, 2019). Aside from general accessibility issues with research-specific STEs, HATs face unique challenges in using a previously validated and tested STE. In particular, the rapid advancement of new AI technologies often leaves STEs behind in integrating autonomous teammates. Further, some commercial AI technologies may not be capable of interfacing

with such context-specific technology. In turn, HATs may struggle to leverage a team-specific STE despite their benefits to measurement and experimentation.

Digital Video Game platforms. Alternatively, video game platforms also present a viable option for HAT research. Trends in online video games and AI technology have yielded many video game platforms where teams and AI are present (Donchin, 1995; Xia et al., 2020; Zeng, 2021). Outside experimental research, these video game platforms serve as a cornerstone of contextualizing and recruiting in HAT research (R. Zhang et al., 2021, 2023). Video game platforms are often heavily contextualized, which can increase the immersion and interaction participants can have in the environment. Further, large companies often handle the development and maintenance of these platforms, reducing the overhead in some research teams. In many cases, these environments can often be highly customizable, allowing them to accommodate specific teamwork experiments better. For example, the ARMA III environment used in prior work contains a highly robust mission editor that allows researchers to place specific objectives and resources for HATs to leverage (see Case Study 1, (Textor et al., 2022)). HAT research's greatest challenge is that many of these environments are not entirely representative of real-world AI applications. In turn, researchers need to be critical in their choice of video game platform to ensure team tasks and interdependence are fundamental to the task. For example, the ARMA III military simulator (Schelble, Lopez, et al., 2022; Textor et al., 2022), Rocket League Soccer Esport (Flathmann, Schelble, Rosopa, et al., 2023; Verhoeven & Preuss, 2020), the Overcooked kitchen management game (Knott et al., 2021; Rosero et al., 2021), and even Minecraft (Amresh et al., 2023; Paleja et al., 2021) have all been identified as beneficial platforms for HAT research. Still, additional platforms can and should be explored by the domain.

Deciding to Use a Physical Research Task

It is important to note that as the prevalence of AI in society grows, so too does the number of people actively working with AI and autonomous technologies. In turn, these populations and environments will continue to inspire the design of

autonomous teammates with a physical presence (Ness et al., 2023; Shaukat et al., 2020), and research will need to inform the designs of these teammates. HATs likely carry unique attributes when in physical contexts, as these teams may need to share the cognitive workload, physical workload, safety, team resilience, and finite resources (Begerowski et al., 2023; Goodrich et al., 2023; Schwind et al., 2018).

However, if a physical environment is justified, conducting research in these environments differs greatly from conducting research in STEs or online surveys. Researchers interested in physical HATs should work on modeling a simulated and repeatable task from a real-world process. For example, prior work has modeled research tasks off of urban search and recovery operations, a common testbed for autonomous systems (Demir et al., 2020; Lyons et al., 2023). However, this process poses a unique challenge to HAT researchers. Specifically, the rapid development of AI and robotics means that many physical environments will likely never be fully controllable by a researcher, and larger companies make many design decisions. Further, experimental results may not always retain relevance, as new robotic systems could vastly change the presentation or capability of a system. Finally, the pricing of these systems is often much more expensive than other physical lab equipment, and many researchers may even find it difficult to access such technologies. As such, it's likely ideal for HAT researchers to leverage digital environments unless their work explicitly calls for physical considerations.

The Utility and Benefit of Prerecorded Survey Tasks

While in-person experiments provide a robust and deep methodology for research, online research studies have become increasingly popular in recent years. In particular, factorial surveys have risen to popularity in AI research due to their ability to present complex AI technology in a safe, low-risk, and low-fidelity environment (Li et al., 2023). Multiple HAT research efforts have leveraged this experimental environment to conduct rapid research efforts (Flathmann, Schelble, McNeese, et al., 2023; Sadeghian et al., 2024; Ulfert et al., 2022; R. Zhang, Flathmann, et al., 2024) (see Case Study 1 and 2 (Lyons et al., 2023; Textor et al.,

2022)). When HAT research is especially concerned with rapidly evaluating autonomous teammates, factorial surveys have been and will be an important tool. Aside from the general challenges that face online factorial surveys, such as making visible and impactful experimental manipulations (Auspurg & Hinz, 2014; Sauer et al., 2020), many important decisions can help make online survey studies a more feasible option for a HAT study. In particular, HAT studies need to accurately communicate the capabilities of autonomous teammates even if humans will not see those capabilities in action (Lyons et al., 2023). Similarly, the roles of human teammates need to be articulated in detail, as humans will not be able to experience those roles and interactions first hand (Flathmann, Schelble, McNeese, et al., 2023).

Synthesizing Recommendations for Research Task Selection

Researchers have numerous options when deciding on a task to leverage within an HAT experiment. In turn, the following list of recommendations has been synthesized from the above text to help quickly guide future research toward an ideal task environment for their experiment:

- Default to using digital task environments unless a research question is explicitly interested in physical aspects of HATs to avoid confounds and improve experimental control.
- When using video game platforms, ensure that the platform can host autonomous teammates, facilitate the formation of team processes, and immerse players in a representative task.
- When using physical platforms, carefully consider how an experiment could generalize to other physical AI technologies.
- Use online survey studies to quickly explore far-future or emerging HAT prototypes.

Deciding on Data Collection

Data collection methodologies round out HAT experiment creation by incorporating the prior fundamental decision points of developing appropriate research questions, deciding the team composition, and designating the needed research environment as the testing platform. Most notably,

selecting these dependent variables ties together one's research question, team composition, and task environment by defining and measuring research question outputs.

Selecting Task-Derived Behavioral Data

It is important to identify the potential data available to HAT experiments and potential methodologies for collecting said data. First and foremost, HAT research that observes task completion, whether synthetic or real, should collect task-related behavioral data. In particular, team effectiveness serves as a core and predominant measure across a variety of HAT studies (Bansal, Wu, et al., 2021; McNeese, Demir, et al., 2021; McNeese et al., 2018). With the advent of autonomous teammates promising significant gains in performance, ignoring performance altogether is likely only helpful for niche research questions and compositions.

The most important consideration in collecting this task data is to ensure that researchers go beyond simple task performance and explore human teammates, autonomous teammates, and whole team performance from multiple perspectives (Harriott & Adams, 2013; McNeese et al., 2018). Indeed, traditional team research highlights how effective teams are described by several emergent states and outcomes, including their shared understanding (Gibson & Cohen, 2003; Mohammed et al., 2010; Salas & Fiore, 2004), situational awareness (Endsley, 2023; Gombolay et al., 2017; Gorman et al., 2011), task performance (Brannick et al., 1997), reward alignment (Sanneman & Shah, 2023), and others. Additionally, HAT experiments would especially benefit from measuring the macrocognitive functions (Miller, 2018; Patterson et al., 2018) of common grounding and coordination (Klein et al., 2005). The utility of behavioral performance and evaluation is especially important to understanding the impact of AI teammates in HATs (illustrated in Case Study 3 (Lyons et al., 2023)). Indeed, the AI's dynamic, complex, and uncertain nature means that evaluation must often come from some observation. Further, the performance maximums of AI technology in isolation can be overwhelming and harmful to HATs (Bansal, Nushi, Kamar, Weld, et al., 2019), meaning this evaluation needs to

happen in actual teams and not simulation. As such, HATs need to be behaviorally evaluated to ensure that AI functions ideally and that their impacts benefit the overall team.

Selecting Quantitative Perceptions of Teammates, Teams, and Tasks

Perception information has also become especially important to HATs, as perceptions may sometimes diverge from similar task-derived considerations. Capturing this perceptual data provides an added layer to autonomous teammates and AI technology evaluation beyond performance. However, where HATs become uniquely challenging is in all of the different perceptions relevant to teams. Indeed, AI teammates need to be perceived positively as interactive technologies, positively as teammates, and even positively as societal shifts. Perceptions of interest to prior work have included performance (Flathmann, McNeese, et al., 2023; McNeese, Schelble, et al., 2021), ethics (Schelble, Lopez, et al., 2022; Textor et al., 2022), emotional intelligence (Mallick et al., 2023), workload (McNeese, Schelble, et al., 2021; Zhang, Raina, et al., 2021), and shared understanding (Schelble, Flathmann, et al., 2022). Finally, many of these perceptions can be measured from different perspectives, such as trust, which has often been measured from multiple dimensions in singular studies (see Case Study 2 (Lyons et al., 2023)). Again, the need for different perceptions is not unique to HATs. Still, HATs present a unique challenge of having an overwhelming number of perceptions simultaneously relevant to autonomous teammates, human teammates, and teams as a whole.

Further, once a perception and correlated measurement tool is chosen, HAT research is challenged to update these measures based on technological innovations. For example, while many trust or technology acceptance metrics have been heavily studied in the automation space (Ghazizadeh et al., 2012; Hoff & Bashir, 2015), the capability, autonomy, and nature of AI and autonomous teammates mean that the content and even nature of these measures likely need to be updated. For example, prior work on ease-of-use has highlighted its importance in accepted automation; however, autonomous teammates are often less reliant on direct use

and more on interdependent collaboration (Flathmann et al., 2024); similarly, recent efforts to explore human-machine interdependence have also expanded upon what is required for humans and autonomy to be interdependent as opposed to humans and automation (Woide et al., 2021). These are just a few examples of how researchers may need to update the nature of these perceptual measures to use them in HAT research.

Including Qualitative Data

Last but not least, qualitative data has served as a foundational component of HAT research over the past decade (Begerowski et al., 2023; R. Zhang, Raina, et al., 2021). Due to the novel existence of AI technology and autonomous teammates, qualitative data has served to identify constructs that experimental research has yet to identify as important to HATs. This affordance can be especially important when one explores real-world tasks, which often see qualitative methodologies (Begerowski et al., 2023) since qualitative data gives participants a more flexible ability to describe their experience. Further, qualitative data can help provide greater fidelity when HAT-specific measurement tools are unavailable.

Mixed-methods research needs to leverage qualitative data to dive deeper into participants' experiences (see Case Study 1 for example (Textor et al., 2022)). In particular, qualitative data provides a robust methodology for participants to describe their experience during an experimental task, which may serve to explain behavioral and performative outcomes observed quantitatively. Within the HAT space, the utility of this data can help triangulate how peoples' personal experiences impacted their behavior within an experiment (Textor et al., 2022). Indeed, autonomous and AI technologies do not exist in vacuums, and the ongoing integration of AI technologies in society continues to change people's expectations and goals for teammates (R. Zhang, Raina, et al., 2021). As such, HAT researchers would likely use various qualitative methods, including grounded theory (Oktay, 2012), phenomenological interviewing (Braun & Clarke, 2012), or others, to expand upon their experimental results and further explore the personal expectations and challenges

humans face when working with AI and autonomous teammates.

Synthesizing Recommendations for Data Collection

Synthesizing the final recommendations provided by this article will help researchers finalize their experimental and methodological design. In turn, the following list of recommendations can be used to improve the quality of data collected in HAT experiments:

- Leverage performance metrics to evaluate autonomous teammates, human teammates, and HATs, but do not solely rely on task performance due to the social and process impacts of autonomous teammates.
- Leverage multiple perception dimensions or tools for the same perception measure, as autonomous teammates can impact social, technological, and procedural outcomes in teams.
- Collect qualitative data to explore nuanced and novel HAT opinions, including those that may be formed through interactions with AI technology outside of your experiment.

Future Research Needs

While this article explores past case studies to present the tools available to HAT experimenters, innovation should not cease in regard to HAT research methodologies. In highlighting the existing research methodologies and decisions available to researchers, this work has also highlighted a number of compromises that researchers have had to make. In turn, this work will conclude by briefly outlining the existing limitations in HAT research methodologies and issuing a call to action to researchers in the improvement of method capabilities within the domain.

Increase the Usage of Real Autonomous Teammates

This article highlights how the approach of operationalizing autonomous teammates through a WoZ methodology, while common, is becoming less relevant when compared to the option of using real

autonomous teammates. Indeed, with years passing since research first showed the potential of real autonomous teammates, continued innovations in AI technology should inspire future HAT experiments (McNeese et al., 2018). Further, while human understanding of AI and autonomous teammates was previously minimal, the rising prevalence of AI in society means that humans are developing a stronger understanding and familiarization of AI technology. Thus, this exposure has the potential to reduce how believable WoZ techniques are to participants, how representative these techniques are of real-world AI, and how relevant perceptions of WoZ autonomous teammates are to real AI technologies. While a number of efforts are working to increase the availability of AI technology in various domains (Alasadi & Baiz, 2023; Hopkins et al., 2023), efforts internal to the HAT research community are needed to increase the knowledge and familiarity of real AI technologies as autonomous teammates. Additionally, continued efforts need to explore how to utilize real AI technology as autonomous teammates while still maintaining control over participant experiences. Compared to WoZ teammates, the general understandability of real AI is less known due to the black-box nature of the technology (Gillani et al., 2023; Rai, 2020). In turn, research efforts that leverage real AI technology carry with them a number of unknowns related to the behavior and rationale of autonomous teammates. From an internal validity standpoint, this can be a challenge to experimental control; however, these unknowns are exactly what real-world people face every day when interacting with AI and autonomous technologies. In turn, prioritizing the incorporation of these AI technologies in future research will better merge research with the real world, bolstering the benefit HATs provide to society.

Develop Shared Synthetic Task Environments

Similar to real AI technologies, further research efforts within the community are needed to either create or standardize synthetic task environments used for empirical experiments. As discussed, research in HATs can leverage various video game environments, such as ARMA III, Minecraft, or Rocket League, as well as multiple in-house

development platforms, such as Blocks World for Teams, CERTT, or NeoCities (Cooke et al., 1999; McNeese et al., 2005; Schelble et al., 2020; Textor et al., 2022). While the latter environments provide decades of validation and metrics that make them highly capable at specific tasks, they often have to be maintained by small teams of researchers, which makes them less customizable and expandable. Meanwhile, video game environments provide an out-of-the-box solution that fosters seamless maintenance and customizability, but their use in research requires intensive task and metric development. In the long term, researchers need to explore the possibility of one or more shared STEs that might provide a common pathway toward research synchronization. While the development of one or more shared platforms would take time, creating platforms explicitly geared toward HAT research with real AI technology will ultimately lead to greater research outcomes. During this development process, multiple contexts and tasks need to be identified, key task and perceptual metrics need to be formed, and collaborations need to happen with real-world teams to ensure external validity. In turn, a greater emphasis needs to be placed on research articles and papers that explicitly contribute toward synthetic task development, and collaborative groups need to be formed with the goal of developing, validating, and sharing these environments.

Create HAT-Specific Quantitative Measurement

While numerous data collection methods and tools retain prevalence in HAT research, limitations still mitigate the impact that HAT research is able to have. While research efforts have continuously leveraged teamwork and technology metrics to evaluate HATs, HATs are continuously highlighting that they are a unique construct that has to be treated as such (McNeese et al., 2023). In turn, the measurement tools used to evaluate HATs also likely need to explicitly measure the values, needs, and definitions of HATs, which is not currently the case. For example, trust in HATs and autonomous teammates represents the evolution of both prior research in automation and teamwork (O'Neill et al., 2022). In turn, the measurement of trust in the field has been pulled from these two domains

(Krausman et al., 2022). However, as the community's understanding of HATs increases, standards and measures need to specify what it actually means for an autonomous teammate to be trusted. In turn, future efforts will better distinguish trust in HATs as the unique construct that it is shaping out to be, which research is already beginning to do (Krausman et al., 2022). Similarly, some constructs in HATs are not even measurable with currently available tools. For example, shared understanding definitions required modification to make them relevant to HATs. Due to the artificial nature of autonomous teammates, the construct itself has mostly been limited to human–human shared understanding (Schelble, Flathmann, et al., 2022), with minimal efforts to create a type of measurable mental model for autonomous teammates (Andrews et al., 2023; Gero et al., 2020). In turn, as future efforts continue to define shared understanding and other constructs in HATs, additional efforts also need to ensure that actual measurement tools are feasible to apply in real-world HATs.

Perform Longitudinal Work

In exploring the research questions within the HAT domain, the successful formation of team constructs serves as a critical component to ensuring HAT success. However, classic teamwork research has continuously shown that these constructs do not form immediately, and it takes time for strong levels of trust, team cognition, and even awareness to form (Li & Roe, 2012; López-Gajardo et al., 2023). However, current research in HATs has been relatively short-term. While multi-hour experiments are not uncommon in the HAT domain (Demir, Likens, et al., 2019; McNeese et al., 2018), little work has explored longitudinal effects over the course of days, weeks, and even months. Most likely, this absence is the combined result of high difficulty in performing longitudinal research and the still early stage of HAT research. Indeed, early HAT efforts have been about providing fundamental knowledge about the novel team type to motivate its future exploration and eventual implementation in the real world. However, as this implementation approaches, the reality of HATs and autonomous teams operating over a longitudinal period also becomes more likely. In turn,

HAT research needs to immediately consider the potential need and role of longitudinal research in the field. This is not to say that all research needs to immediately shift to longitudinal studies; however, existing research on team constructs in HATs should consider longitudinal effects and extensions to ongoing research.

Limitations and Future HAT Research Innovations

While this article can posit multiple important HAT experiment considerations, it is worth acknowledging that no document can fully explore every decision researchers need to make when creating and performing HAT experiments. Notably, this article represents four decisions that exist across nearly all HAT experiments. Aside from these four considerations, researchers may also encounter challenges in the actual design of a real or WoZ AI teammate, the creation of a new synthetic task environment, the use of physiological data, or other considerations. Moving forward, broader reviews of the domain should be encouraged to better identify the totality of decisions researchers need to make, and additional meta-contributions will be needed to guide researchers through these decisions. Due to the fundamental nature of this article, the HAT domain will still need systematic reviews of different methods, measurements, and decisions in HAT research. Notably, each decision point discussed in this article could likely serve as its own systematic review. Ideally, this review will provide a starting place and structure to perform these future reviews and further define the HAT domain and its methods.

Conclusion

Within the current state of HAT research, significant strides are being made to design, observe, and measure HATs through experiments. However, with the rapid progression of the research field and its derivation from multiple research fields, researchers may often find themselves overwhelmed and at a loss. Key insights and recommendations have been made to guide research question formation, team composition selection, research task selection, and data

collection procedures. To succinctly aid future researchers, these recommendations have also been compiled and organized within [Table 1](#). Further, this robust exploration of experimental considerations in HAT research has highlighted current methodological limitations in the field, which have been synthesized to motivate future innovation. In contributing to this article, researchers will gain assurance and momentum in their experimental efforts, ultimately increasing the impact of their individual research and the HAT domain as a whole.

Appendix

Case Study 1: Exploring the relationship Between Ethics and Trust in Human–Artificial Intelligence Teaming: A Mixed-Methods Approach

Research Questions. The first case study for this work comes from Textor and colleagues ([Textor et al., 2022](#)). Case Study 1 is a bit unique, as it presents two studies, an experiment and a follow-up qualitative study used to contextualize experimental

Table 1. A Full List of Recommendations This Article has Derived From High-Impact Experiments in the HAT Domain.

Synthesized Recommendations for Designing High-Impact HAT Experiments	
Decision point	Recommendation
Research question development	Explore prior research in teamwork and technology to identify inputs, processes, and outputs of interest to HATs, but identify how those factors are likely to change due to increased levels of autonomy, AI complexity, and teamwork found in HATs View HAT processes and outputs as potential inputs for research questions, and also view them as inputs for the design and parameters of AI and autonomous teammates Leverage HAT research questions to explore novel teamwork processes in HATs, recontextualize existing teaming factors in HATs, or evaluate autonomous teammate technologies
Team composition selection	To prevent autonomous teammates from overwhelming HATs, use the minimum required number of autonomous teammates in an HAT to answer your research question If a research question is not explicitly concerned with physical autonomous teammates, consider the use of digital autonomous teammates to reduce confounds with robotic design When possible, use real AI teammates to increase the applicability of research to modern HATs, but remember that a WoZ can be a fall-back if an AI is not available or possible Consider having multiple humans to explore human–human relationships in HATs, but only use confederates when necessary to more accurately explore these relationships
Task and platform selection	When using video game platforms, ensure that the platform can host autonomous teammates, facilitate the formation of team processes, and immerse players in a representative task When using physical platforms, carefully consider how an experiment could generalize to other physical AI technologies Use online survey studies to explore far-future or emerging HAT prototypes quickly
Data collection selection	Leverage performance metrics to evaluate autonomous teammates, human teammates, and HATs, but do not solely rely on task performance due to the social and process impacts of autonomous teammates Leverage multiple perception dimensions or tools for the same perception measure, as autonomous teammates can impact social, technological, and procedural outcomes in teams Collect qualitative data to explore nuanced and novel HAT opinions, including those that may be formed through interactions with AI technology outside of your experiment

results (research questions listed in Table 2). From an input perspective, Case Study 1’s RQ was concerned with the actions and decisions autonomous teammates make in HATs, and the output perspective highlights the perceived ethicality and trust that human teammates form for their AI teammates. While implicit, the RQs are also concerned with interaction and perception formation processes that human teammates undergo when observing these actions. While these RQs share overlap with other domains, within the HAT domain, this research question merits unique considerations due to autonomous teammates’ challenges in making ethical decisions and the uncertainty that surrounds AI and autonomous technologies.

Team Composition. For composition (Textor et al., 2022), Textor and colleagues leveraged a team

where humans and autonomous teammates worked toward a shared goal by making joint ethical decisions informed by interdependent data collection efforts. Within this composition, the autonomous teammates leveraged were either a human acting as an AI (WoZ) or simulated text that was presented as coming from an AI. These two approaches are common within HAT and other domains, yet their operationalization is unique given the increasingly blurred line between autonomous and human teammate communication and capability (Brandizzi, 2023) In addition to autonomous teammates, Case Study 1 included two human teammates who were participants with different roles, allowing for multiple human perspectives to answer the posited research question. Within HATs, these two perspectives can be critical to experiments, as different team roles may be privy to different observations, transparencies, and desires of autonomous teammates. Merging these decisions in team composition, Case Study 2 explored different autonomous teammates, different human perspectives, and human–human relationships through these team compositions (Table 3).

Research Task. Case Study 1 used two research tasks: an online survey-based command and control task and an in-person experiment in a video

Table 2. Research Questions Explicitly Stated in Case Study 1.

Case Study 1 research questions
RQ1 Do people assign ethicality to an [autonomous Teammate]’s actions?
RQ2 If so, do their perceptions of an [autonomous Teammate]’s decisions or actions affect their trust in that AT?

Table 3. Team Composition Details for Case Study 1.

Composition consideration	Case Study 1 selection
Study 1—experiment	
Team size	Two human parties and one autonomous teammate
Human roles	Environment navigation and conflict navigation
Human confederates	No
Autonomous teammate roles	Decision-making
Autonomous teammate type	Simulated text
Modality	Digital
Study 2—qualitative follow-up	
Team size	Two human teammates & one autonomous teammate
Human roles	Environment navigation & information gathering
Human confederates	No (excluding Wizard of Oz)
Autonomous teammate roles	Decision-making
Autonomous teammate type	Wizard of Oz
Modality	Digital
RQ reminder	
RQ1: Do people assign ethicality to an AT’s actions?	
RQ2: If so, do their perceptions of an AT’s decisions or actions affect their trust in that AT?	

game platform (detailed in Table 4). Based on the research questions and team composition selected, the online survey was constructed based on prior research in military teams, and the ARMA III platform served to derive qualitative data and support the second in-person study. Starting with the survey, an online experiment was created to explore HAT decision-making surrounding military actions with ethical implications (research question 1). For ARMA III, Case Study 1 created supportive qualitative data and contextualized the importance of ethics and trust in military

operations (research question 2). Case Study 1 highlights the diversity of research contexts available to HATs. As AI technologies become more visible and collaborative, online surveys can depict and even facilitate interaction with real AI. Similarly, commercial video game platforms are gaining increased fidelity and interactivity, making them closer to simulations of the real world.

Data Collection. Closing this article’s analysis of Case Study 1, Textor and colleagues carry their research questions regarding the inclusion of trust and perceived ethicality into their data collection. Using behavioral, perceptual, and qualitative data, Case Study 1’s research question is answered through multiple data sources. The trust participants formed for their autonomous teammates was measured starting with the focus on the case study’s research question. Further, while experimental manipulations changed the ethicality of the autonomous teammate actions, researchers still chose to measure the perceived ethicality human teammates formed. However, one of the most impactful decisions within Case Study 1 is that Textor and colleagues chose to take a mixed-methods approach (detailed in Table 5). While mixed-methods approaches are not unique to HATs, Case Study 1 highlights how these approaches can uniquely benefit HATs, as quantitative data can be used for autonomous evaluation while qualitative data can improve the future design of autonomous teammates.

Table 4. Task and Platform Details for Case Study 1.

Task consideration	Case Study 1 details
Study 1—experiment	
Modality	Digital
Delivery	Online
Platform type	Factorial survey
Platform choice	Text scenarios
Task goals	Military course of action planning
Study 2—qualitative follow-up	
Modality	Digital
Delivery	In-person
Platform type	Adapted video game
Specific choice	ARMA III military simulation
Task goals	Military course of action planning
RQ reminder	
RQ1: Do people assign ethicality to an AT’s actions?	
RQ2: If so, do their perceptions of an AT’s decisions or actions affect their trust in that AT?	

Table 5. Data Collection Details From Case Study 1.

Data consideration	Case Study 1 details
Data focus	Perceived ethics and trust
Quantitative task data	None
Quantitative perceptions	Autonomous teammate ethicality, agreement with autonomous teammate, trust in autonomous teammate
Measurement tools used	Single item measures derived from (J. Lee & Moray, 1992; J. D. Lee & Moray, 1994; Walliser et al., 2016)
Analysis tools used	Mixed effects ANOVA
Qualitative data	Yes (focus groups)
Qualitative methodology used	Thematic analysis (Braun & Clarke, 2012)
RQ reminder	
RQ1: Do people assign ethicality to an AT’s actions?	
RQ2: If so, do their perceptions of an AT’s decisions or actions affect their trust in that AT?	

Case Study 2: Explanations and Trust: What Happens to Trust When a Robot Partner Does Something Unexpected?

Research Questions. This article’s second case study refers to an article by Lyons and colleagues (Lyons et al., 2023). While the article does not explicitly state its research question, both the title and introduction of the piece see an inquisitive interest in the relationship between unexpected actions, trust, and explanations (research questions listed in Table 6). Exploring Case Study 2’s RQs yields a clear definition of the inputs and outputs the authors are interested in exploring. Similar to Case Study 1, Lyons and colleagues are especially interested in the behavioral actions taken by autonomous teammates and their impact on trust. However, Case Study 2 was more concerned with the unexpected nature of the autonomous teammate’s actions, and this change likely impacts the ideal tasks that would be used to answer this research question. Again, trust and unexpected actions are not inherently unique to HAT research, as automation and general teamwork would see similar needs. However, this Case Study 2 highlights how trust changes with autonomous teammates as they are more complex and less

transparent than automation, as AI technologies are inherently black-box and teammates require an inherent level of autonomy to perform actions that may not be expected; in turn, how one explores and evaluates the perception of autonomous teammates, unexplained actions, and trust has to hold a different perspective.

Team Composition. In Case Study 2, Lyons and colleagues take a different approach to team composition (summarized in Table 7) (Lyons et al., 2023). Participants in their experiment were not a part of a HAT composition. Rather, their experiment had participants observe recordings of dyadic human–robot teams. This methodology and composition provided specific affordances in answering their research question. Specifically, the most HAT-specific affordance was that they were able to depict a specific robotic teammate without having that robot be capable of real-time teaming. If this study had been performed in person, then the autonomous teammate in question would have needed real-time capabilities. Second, by recording a human teaming with this robot, they could control the human teammate’s expectations, making it much easier to depict the autonomous teammate acting counter to those

Table 6. Research Questions Derived From Case Study 2.

Case Study 2 research questions	
RQ1	How are perceptions of trust and trustworthiness in human–autonomy teams impacted by autonomous teammates performing unexpected behaviors?
RQ2	How do explanations impact the trust humans form for autonomous teammates performing unexpected behaviors?

Table 7. Team Composition Details for Case Study 2.

Composition consideration	Case Study 2 selection
Team size	One human teammate and one autonomous teammate
Human roles	Decision-making
Human confederates	Yes
Autonomous teammate roles	Information gathering
Autonomous teammate type	Prerecorded robot
Modality	Physical (prerecorded)
RQ reminder	
RQ1: How are perceptions of trust and trustworthiness in human–autonomy teams impacted by autonomous teammates performing unexpected behaviors?	
RQ2: How do explanations impact the trust humans form for autonomous teammates performing unexpected behaviors?	

expectations. This choice allowed the research study to focus solely on observing unexpected behaviors rather than trying to create a scenario where the participant felt that their expectations were violated.

Research Task. Unlike Case Studies 1 and 3, Lyons and colleagues leveraged a physical task environment for their study. While their environment was presented via a prerecorded video, participants observed autonomous teammates interacting with a physical environment (detailed in Table 8). Due to prioritizing a real-world physical environment, Case Study 2’s task was inspired by a desire to understand what human–AI teamwork would resemble with real-world robotic systems. Specifically, Case Study 2’s task is motivated by real-world search and rescue tasks, where the use of autonomous robotic teammates has been continuously explored due to their minimization of safety concerns and increased sensing capabilities (N. Cooke et al., 2020; Nguyen & Gonzalez, 2023). However, despite using a physically embodied autonomous teammate to complete a common physical task, Case Study 2 still had participants interact through a digital modality. This choice enabled Case Study 2 to conduct a physical HAT experiment in a safe, low-cost, and accessible environment, which may not always guaranteed when performing HAT research with real physical autonomous teammates.

Data Collection. For Case Study 2, Lyons and colleagues take a purely quantitative approach to exploring the trust humans form for their autonomous robot partners. In particular, this Case Study’s research question centers around the

construct of trust as the central process and output of interest. Given this task’s online and prerecorded nature, trust was only explored perceptually, as participants did not have any behavioral metrics to record. However, one of Lyons and colleagues’ more notable decisions was to explore perceptual trust from multiple perspectives (shown in Table 9). This expansion of trust and other measures is especially important to HATs due to the social, functional, and collaborative capabilities AI requires to be autonomous teammates.

Case Study 3: Teaming With a Synthetic Teammate: Insights Into Human-Autonomy Teaming

Research Questions. The final case study comes from McNeese and colleagues (N. J. McNeese et al., 2018). Case Study 3 experimentally compares HATs to all-human teams to provide the first empirical validation of HATs performing similarly or better to all-human teams (research questions listed in Table 6). Case Study 3 is much more focused on the underlying structure and composition of teams as an input and how such structures influence a team’s performance capabilities as an output. In turn, Case Study 3’s research question explores how performance-related processes and outputs may differ in HATs. Thus, the eventual focus of experimental and methodological decisions must accommodate and explore these processes’ differences. Despite the seemingly simple and fundamental nature of this research question compared to Case Studies 1 and 2, critical decisions are still needed when selecting team compositions, experimental tasks, and dependent

Table 8. Task and Platform Details for Case Study 2.

Task consideration	Case Study 2 details
Modality	Digital
Delivery	Online
Platform type	Factorial survey
Platform choice	Video scenarios
Task goals	Urban search and recovery robotics
RQ reminder	
RQ1: How are perceptions of trust and trustworthiness in human-autonomy teams impacted by autonomous teammates performing unexpected behaviors?	
RQ2: How do explanations impact the trust humans form for autonomous teammates performing unexpected behaviors?	

Table 9. Data Collection Details From Case Study 2.

Data consideration	Case Study 2 details
Data focus	Trust
Quantitative task data	None
Quantitative perceptions	Trust in autonomous teammate, trustworthiness of autonomous teammate (ability, benevolence, integrity), responsibility attribution
Measurement tools used	Both scales and single-item measures were used and taken from (Lyons & Guznov, 2019; Mayer et al., 1995)
Analysis tools used	Mixed effects ANOVA
Qualitative data	None
RQ reminder	
	RQ1: How are perceptions of trust and trustworthiness in human–autonomy teams impacted by autonomous teammates performing unexpected behaviors?
	RQ2: How do explanations impact the trust humans form for autonomous teammates performing unexpected behaviors?

variables to ensure the empirical answer created for this question is still valid. (Table 10)

Team Composition. Finally, Case Study 3 from McNeese and colleagues explores multiple team compositions (see Table 11) (N. J. McNeese et al., 2018). In particular, Case Study 3 explores an all-human team composition, an HAT with a real autonomous teammate, and an HAT with a WoZ teammate. A unique aspect of Case Study 3 is that it leverages both a WoZ and a real autonomous teammate technique, as most studies explore one or the other. In turn, Case Study 3 compared the performance capabilities of two different types of HATs to a human–human team benchmark. This utilization of multiple comparative compositions provides a unique benefit to HAT research. In particular, HAT research can leverage a WoZ to explore the potential of fully fledged autonomous teammates and a real AI or autonomous technology to explore the current state of autonomous teammates. Leveraging both allowed Case Study 3 to effectively compare human–human performance to HATs at two different time scales; modern HATs and far-future HATs. This compositional choice enhanced the depth of Case Study 3’s answer to their research question.

Research Task. Lastly, McNeese and colleagues used an STE specifically designed to evaluate teamwork. The specific research questions, the composition needs, and the overall goals of Case Study 3 led the authors to select the platform and task outlined in Table 12. For Case Study 3, the CERTT UAV STE

Table 10. Research Questions Derived From Case Study 3.

Case Study 3 research questions
RQ1 Do human–autonomy teams have the ability to outperform all-human teams?

provided a fundamental research environment since it has a foundation in measuring team performance (Cooke & Shope, 2004; Cooke & Shope, 2005). This was critical to Case Study 3, as the goal was to provide a foundational experiment that differentiated HAT performance from prior teams, so using a validated teaming environment was necessary. Additionally, at the time, the CERTT UAV STE was one of the only platforms that had a working and validated real autonomous teammate (Ball et al., 2010), which was needed since the case study needed to leverage both a WoZ and real autonomous teammate to answer its research question. As such, Case Study 3 directly benefited from using this validated and high-fidelity STE. However, these environments are often less accessible than surveys or gaming environments, and very few HAT-specific STEs exist today. In turn, the selection and use of HAT-specific STEs may not be available to all HAT researchers.

Data Collection. Concluding the exploration of Case Study 3, McNeese and colleagues utilize a quantitative approach to explore their research questions on the performance capabilities of HATs. However, unlike Case Study 2, they do so entirely with behavioral metrics rather than

Table 11. Team Composition Details for Case Study 3.

Composition consideration	Case study 3 selection
Team size 1	Three human teammates
Team size 2	Two human teammates and one autonomous teammate (real AI)
Team size 3	Two human teammates and one autonomous teammate (VVoZ)
Human roles	Environment navigation, information gathering and decision-making (when three-human team)
Human confederates	No (excluding Wizard of Oz)
Autonomous teammate roles	Decision-making
Autonomous teammate type	Real AI and Wizard of Oz
Modality	Digital
RQ reminder	RQ1: Do human–autonomy teams have the ability to outperform all-human teams?

Table 12. Task and Platform Details for Case Study 3.

Task consideration	Case Study 3 details
Modality	Digital
Delivery	In-person
Platform type	STE
Platform choice	Cognitive engineering research on team tasks UAV STE
Task goals	Unmanned aerial vehicle piloting
RQ reminder	RQ1: Do human-autonomy teams have the ability to outperform all-human teams?

Table 13. Data Collection Details From Case Study 3.

Data consideration	Case Study 3 details
Data focus	Team performance
Quantitative task data	None
Quantitative perceptions	Task performance, processing efficiency, team process ratings, verbal behavior ratings, situation awareness
Measurement tools used	Task completion efficiency, roadblock resolution, and team communication (all task derived)
Analysis tools used	ANOVA
Qualitative data	None
RQ reminder	RQ1: Do human–autonomy teams have the ability to outperform all-human teams?

perceptual metrics to answer their research question. In answering this question, Case Study 3 highlights an important aspect of behavioral measures in HATs: autonomous and AI teammate behavior can play a key role in measuring and modeling team performance. Indeed, the often black-bock and nondeterministic nature of

AI teammates means that they often need to be evaluated behaviorally alongside their human teammates. Study 3 tackles this challenge through a multi-measurement view of performance, which looks at the benefits these teammates can have to team processes and overall task performance (Table 13).

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